English

(1001CJA102922058)

Test Pattern

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
PART TEST
04-12-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS, TOAS, TAAS, TNAS, I(A) & LIVE-I(A)] (SCORE-I)

Time : 3 Hours **Maximum Marks : 186**

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble ◯ **COMPLETELY.**
12. The correct way of darkening a bubble is as : ●
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **g = 10 m/s²** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS	
Atomic No. :	H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58
Atomic masses :	H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca=40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba=137, Ce = 140

• **Boltzmann constant**	$k = 1.38 \times 10^{-23}\ JK^{-1}$
• **Coulomb's law constant**	$\frac{1}{4\pi\varepsilon_0} = 9\times10^9$
• **Universal gravitational constant**	$G = 6.67259 \times 10^{-11}\ N\text{–}m^2\ kg^{-2}$
• **Speed of light in vacuum**	$c = 3 \times 10^8\ ms^{-1}$
• **Stefan–Boltzmann constant**	$\sigma = 5.67 \times 10^{-8}\ Wm^{-2}\text{–}K^{-4}$
• **Wien's displacement law constant**	$b = 2.89 \times 10^{-3}\ m\text{–}K$
• **Permeability of vacuum**	$\mu_0 = 4\pi \times 10^{-7}\ NA^{-2}$
• **Permittivity of vacuum**	$\in_0 = \frac{1}{\mu_0 c^2}$
• **Planck constant**	$h = 6.63 \times 10^{-34}\ J\text{–}s$

Space for Rough Work

HAVE CONTROL ⟶ HAVE PATIENCE ⟶ HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 32)

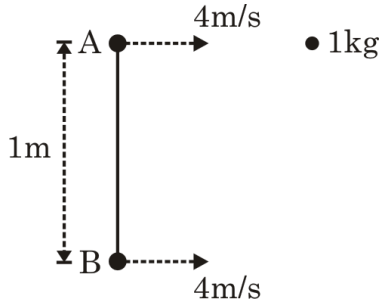
- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : –1 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –1 marks.

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1. Two points masses A and B each of 1kg joined by 1m rod (light) translates on horizontal surface as shown in figure. A particle of mass 1 kg at rest on horizontal surface sticks to ball as it collides with ball A. then :



(A) Linear speed of a ball A just after collision is 2m/s

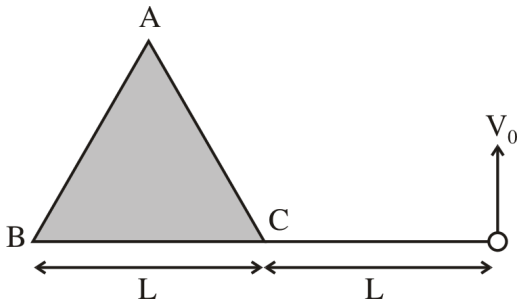
(B) Linear speed of ball B just after collision is 4 m/s

(C) Velocity of centre of mass of system is 8/3 m/s

(D) Angular speed of system, about centre of mass after collision 2 rad/s.

Space for Rough Work

2. As shown in the figure, ABC is a triangular plate of mass M and side length L each. Its mass distribution is non-uniform but its centre of mass is at the centroid. Moment of inertia of this plate about an axis passing through its centroid and perpendicular to its plane is given by $\frac{ML^2}{24}$. End C of the plate is connected to a massless and inextensible string of length L as shown. At the other end of the thread a point particle of mass M is connected. At t = 0 the point mass is given initial velocity of v_0 perpendicular to the length of the string. Choose the correct option(s).



(A) The radius of curvature of the path of the point mass just after projection is 3L.

(B) The radius of curvature of the path of the point mass just after projection is 4L.

(C) The tension in the string just after projection is $\frac{Mv_0^2}{4L}$

(D) Net acceleration of end B and C of plate just after the projection will be same in magnitude.

3. A point object starts moving in a circle of radius 1 mm with constant angular acceleration of 1 rad/s^2 in a plane perpendicular to the principal axis at a distance of 20 cm from the pole of a concave mirror of focal length 25 cm. At time t = 2s after the motion starts

(A) The ratio of acceleration of image and the object is $\sqrt{29}$.

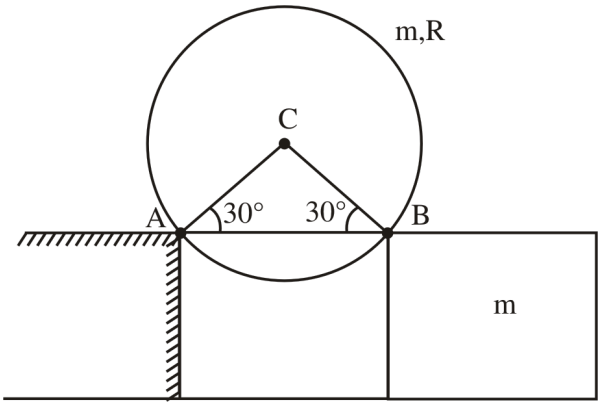
(B) The ratio of acceleration of image and the object is 5.

(C) Magnitude of relative velocity of object with respect to image is 8 mm/s

(D) Magnitude of relative velocity of object with respect to image is 12 mm/s

Space for Rough Work

4. A uniform circular ring of mass m = 5kg and radius R = 30 cm is hinged at the point A of a stationary surface with the line joining point A to the centre C of the ring is initially vertical. The ring is also in contact with the Plank of mass m = 5 kg at point B which is placed on a smooth horizontal surface. Initially points A and B are almost touching and the line AC is vertical and the system is released from rest. Then the velocity of the Plank when the line AC makes angle 30° with the horizontal as shown will be (there is no friction between the ring and the plank and g = 10 m/s^2)



(A) 1 m/s (B) 2 m/s (C) 3 m/s (D) 4 m/s

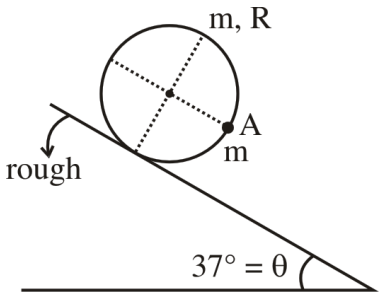
5. A ray of light moving along vector $(5\sqrt{2}\hat{i} - 5\hat{j} - 5\hat{k})$ undergoes refraction at an interface of two media, which is y-z plane. The refractive index for $x \leq 0$ is $\sqrt{3}$ while for $x \geq 0$ it is $\sqrt{2}$. Then the unit vector along the refracted ray is :

(A) $\frac{\sqrt{3}\hat{i} - \sqrt{2}\hat{j} - \sqrt{2}\hat{k}}{\sqrt{8}}$ (B) $\frac{\sqrt{3}\hat{i} - \sqrt{3}\hat{j} - \sqrt{2}\hat{k}}{\sqrt{8}}$

(C) $\frac{\sqrt{2}\hat{i} - \sqrt{3}\hat{j} - \sqrt{3}\hat{k}}{\sqrt{8}}$ (D) $\frac{\sqrt{2}\hat{i} + \sqrt{3}\hat{j} - \sqrt{3}\hat{k}}{\sqrt{8}}$

Space for Rough Work

6. A particle of mass m = 1 kg is rigidly attached to the circumference at point A of a uniform circular ring of mass m = 1kg and radius R = 20 cm placed on a rough inclined plane of inclination $\theta = 37°$ as shown in the figure. Then choose the correct option(s) just after the ring is released from rest. (there is no slipping between the ring and inclined plane and $g = 10\ m/s^2$)



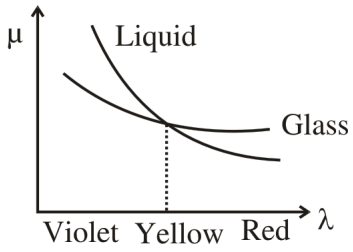
(A) The frictional force acting on the ring due to inclined plane is 4N.

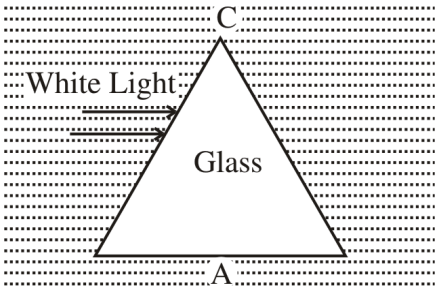
(B) The frictional force acting on the ring due to inclined plane is 2N.

(C) The normal force acting on the ring due to inclined plane is 22N.

(D) The normal force acting on the ring due to inclined plane is 11N.

7. A glass prism is immersed in a hypothetical liquid. The graph shows the variation of refractive index of glass and liquid with the wavelength λ of light. When a ray of white light is incident parallel to the base of the prism, then select the correct option





(A) Yellow ray travels without deviation.

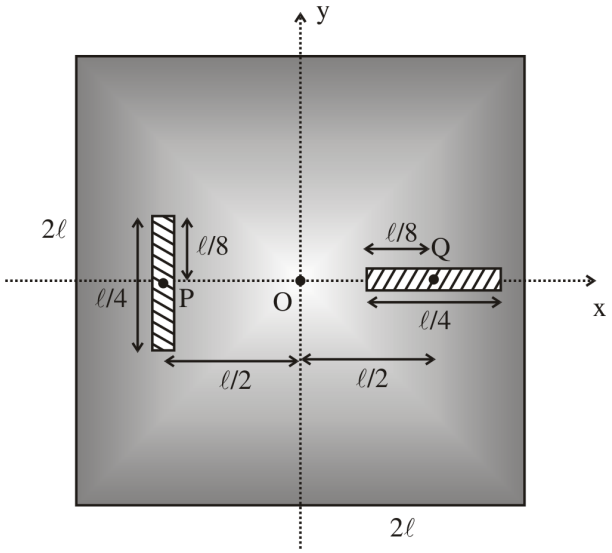
(B) Blue ray is deviated towards vertex.

(C) Red ray is deviated towards base.

(D) There is no dispersion.

Space for Rough Work

8. Two rods each or mass m and length $\frac{\ell}{4}$ are hinged on a larger square plate of same mass and edge length 2ℓ. Both rods can freely rotate about hinge points P and Q in the plane of square plate. Choose the correct option(s).



(A) The moment of inertia of the system about x-axis is $\frac{65}{192}m\ell^2$.

(B) The moment of inertia of the system about y-axis is $\frac{161}{192}m\ell^2$.

(C) The moment of inertia of the system about z-axis is $\frac{226}{192}m\ell^2$.

(D) The moment of inertia of the system about z-axis is $\frac{224}{192}m\ell^2$.

Space for Rough Work

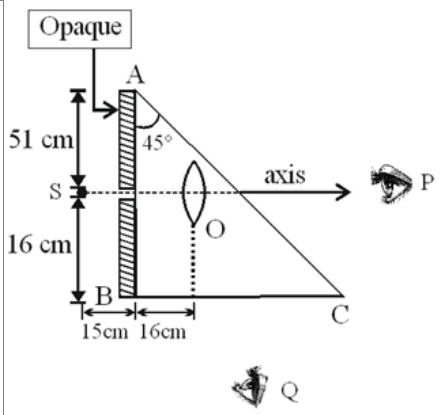
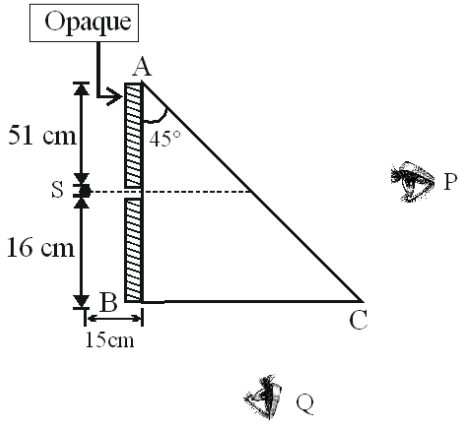
SECTION-I (ii) : (Maximum Marks: 12)

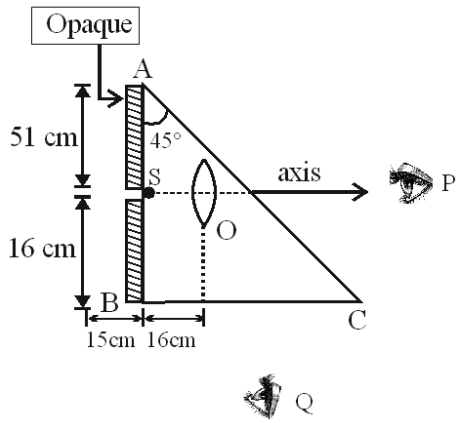
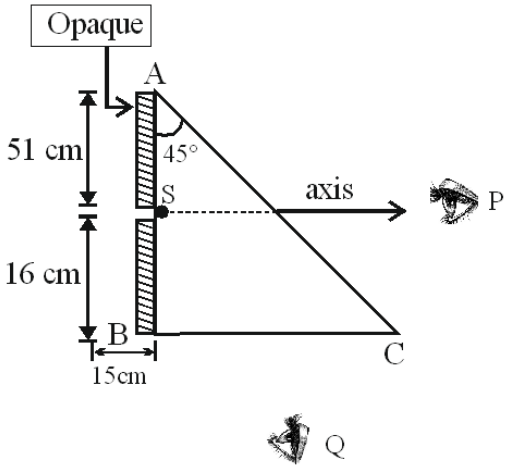
- This section contains **TWO (02)** List-Match sets.
- Each List-Match set has **TWO** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has Four entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **Four** options are given in each Multiple Choice Question based on List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks	:	+3	If ONLY the option corresponding to the correct combination is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e., the question is unanswered)
Negative Marks	:	–1	In all other cases

Paragraph for Questions 9 and 10

The figures show an isosceles right angled prism of ($\mu = 1.6$) and there is a point source situated either inside or outside of the prism also we have two observers P & Q. Opaque sheet having a small hole lies outside the prism. List-I represents location of observers and List-II represents the nature & location of images.

	List-I		List-II
(I)	 A cavity is made inside the prism in the shape of equi-convex lens of radius of curvature 20 cm and the cavity is filled with a transparent liquid of refractive index $\mu = 2.4$. Source is outside the prism as shown	(P)	Only P can see the image
(II)	 Source is outside the prism as shown and there no any cavity inside the prism.	(Q)	Only Q can see the image

(III)	 A cavity is made inside the prism in the shape of equi-convex lens of radius of curvature 20 cm and the cavity is filled with a transparent liquid of refractive index $\mu = 2.4$. Source is inside the prism as shown	(R)	Image is virtual located at 55/8 cm above the surface BC
(IV)	 Source is inside the prism as shown and there is no cavity inside the prism.	(S)	Image is real
		(T)	No observer can see the image
		(U)	Both (P) & (Q) can see the image

9. Which of following is correct combination if medium outside the prism have refractive index $\mu = 1$.

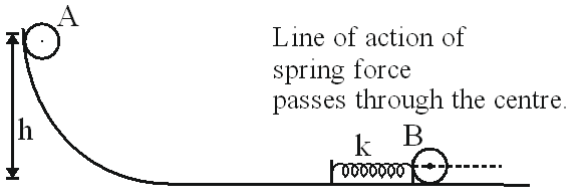
(A) I – R (B) II – R (C) I – P (D) III – P

10. Which of following is correct combination if outside medium is $\mu = 1.6$.

(A) I – S (B) II – R (C) III – S (D) IV – Q

Space for Rough Work

Paragraph for Questions 11 and 12



The figure shows two identical rollable objects held at rest and the spring shown is unstretched and unattached to any object. The complete surface is sufficiently rough to sustain pure rolling all the time. Mass of each object is m and radius r is very small compared to the height h. Spring constant of spring is k. The object kept at height is released so that it goes on to compress the spring. Possible rollable objects are given in List-I. Some values are given in List-II.

	List-I		List-II
(I)	Solid sphere	(P)	$\sqrt{\frac{3m}{4k}}$
(II)	Hollow sphere	(Q)	$\sqrt{\frac{3m}{2k}}$
(III)	Ring	(R)	$\sqrt{\frac{m}{k}}$
(IV)	Disc	(S)	$\sqrt{\frac{5m}{3k}}$
		(T)	$\sqrt{\frac{5m}{6k}}$
		(U)	$\sqrt{\frac{7m}{10k}}$

11. The correct match for the ratio of maximum compression in spring to maximum speed of the object A is

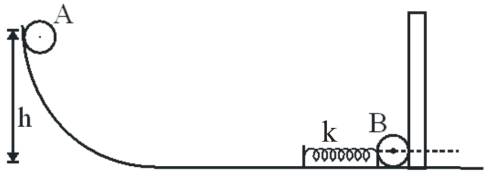
(A) I-P, II-R, III-U, IV-S

(B) I-R, II-U, III-S, IV-T

(C) I-U, II-T, III-R, IV-P

(D) I-S, II-R, III-Q, IV-U

12. If a smooth fixed vertical wall exists to the right of the lower object touching it. The correct match for the ratio $\frac{\text{maximum compression in spring}}{\sqrt{2}(\text{maximum speed of the objective A})}$ is



(A) I–T, II–P, III–S, IV–Q

(B) I–Q, II–S, III–P, IV–T

(C) I–R, II–Q, III–P, IV–S

(D) I–U, II–T, III–R, IV–P

Space for Rough Work

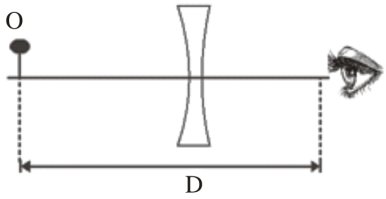
SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

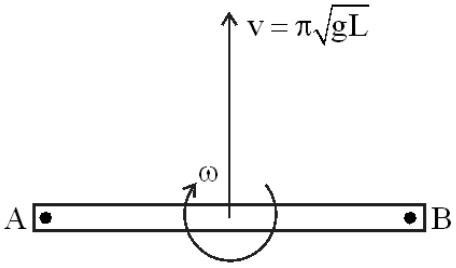
Full Marks : +3 If ONLY the correct numerical value is entered.
Zero Marks : 0 In all other cases.

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1. O is a small object placed at a distance D from eye of an observer. A concave lens of focal length f (f < D) is placed near to eye and image of the object is viewed. Now the lens is moved towards object O away from the eye. Find the distance (in m) of lens from object when appeared size of image is smallest (D = 6m, f = 20 cm)

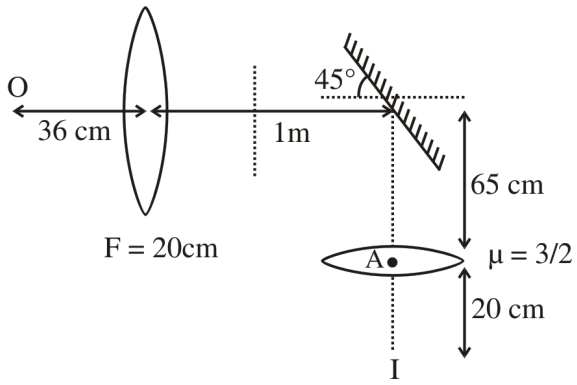


2. Bahubali tossed a rod of length L = 10m from horizontal position at t = 0 such that its centre receives a velocity $v = \pi\sqrt{gL}$ in vertically upward direction and some angular velocity about its centre. The rod lands back to its point of projection in horizontal position. During its course of flight its angular velocity remains constant and it made one complete rotation. Calculate the time (in sec) after which end B is at its highest vertical position given that rod rotates in vertical plane. Take solution of equation $\cos\theta = 2\theta$ to be $\theta = 0.45$ and $\sin(0.45) = 0.43$ if needed.



Space for Rough Work

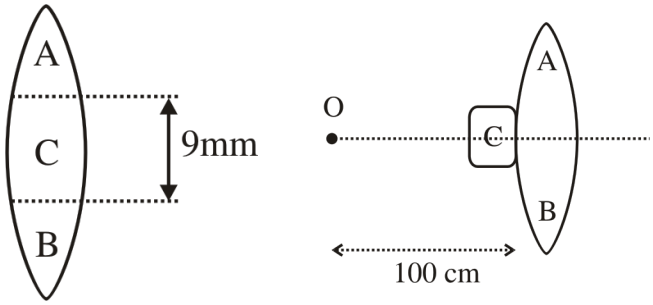
3. The final image I of the object O shown in the arrangement is formed at point 20 cm below a thin equi-convex lens, which is at a depth of 65 cm from principal axis. Calculate the radius of curvature in cm of lens kept at A.



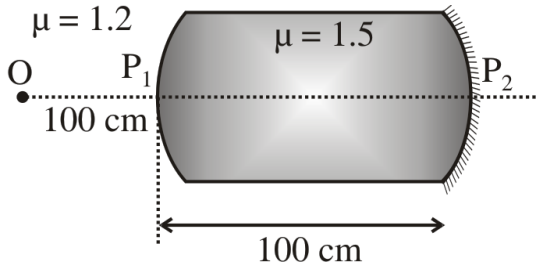
4. In a setup of displacement method experiment, distance between the screen and a light source is 100 cm and the lens used has a small aperture. By moving the lens between the source and the screen, sharp images are obtained on the screen for two different positions of the lens separated by a distance x cm. The ratio of sizes of these two images is 9 : 1. The focal length of the lens used is f cm. The value x is :

Space for Rough Work

5. A thin convex lens of focal length 50 cm is cut into three parts A, B and C by the planes parallel to its optic axis. The thickness of the middle layer C is 9 mm. The middle layer is now removed and the two parts A and B are stuck together. Then the part C is also placed in contact of this lens symmetrically as shown in the figure. Now, a point object O is placed at a distance 100 cm left on the optic axis of the part C. Find the distance (in mm) between the two images formed by the parts A and B.



6. A luminous point object O is placed in front of spherical glass of radius R = 20 cm. The final real image is formed after refraction through the spherical surface, followed by reflection at the concave mirror of radius R = 400 cm and second refraction through the spherical surface. Find the distance (in cm) of this final image from the pole P_1 of the spherical surface.



Space for Rough Work

PART-2 : CHEMISTRY

SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

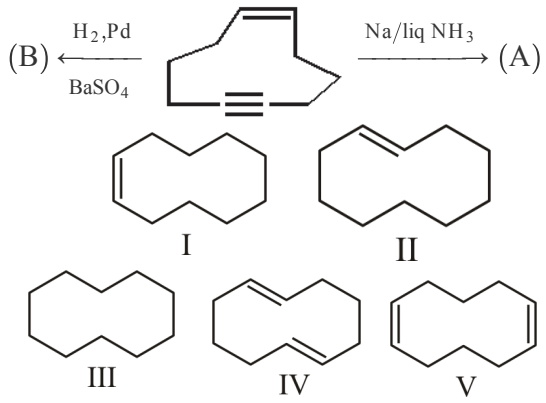
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : –1 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –1 marks.

1.



Select the correct statement(s) regarding the product A & B.

(A) B is V

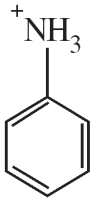
(B) A is IV

(C) A and B both are III

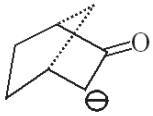
(D) A is II and B is I

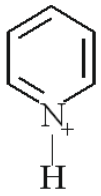
Space for Rough Work

2. In which of the following species, the positive or negative charge is involved in resonance.

(A) 

(B) $H_3C-\overset{\oplus}{C}=O$

(C) 

(D) 

3. $\underset{(C)}{C_8H_{14}} \xrightarrow[(leqvt)]{H_2/Pd} \underset{(D)}{C_8H_{16}} \xrightarrow[Zn]{O_3} \underset{(only\ product\ formed)}{2-\text{bu tan one}}$

Select the correct statement(s) regarding above reaction sequence.

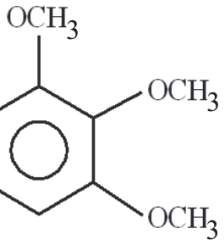
(A) Compound 'C' may be an alkyne or a conjugated diene.

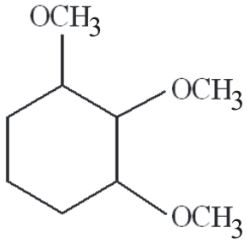
(B) Compound 'D' shows markonikov addition with HBr & anti markonikov addition with HBr/ROOR.

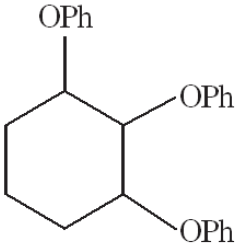
(C) 2-butanone has no chain isomers.

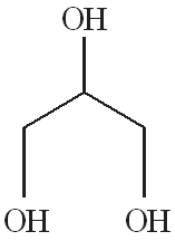
(D) Compound 'D' shows geometrical isomerism but doesn't show optical isomerism.

4. One mole of the following compound(s) can consume more than 4 moles of HI.

(A) 

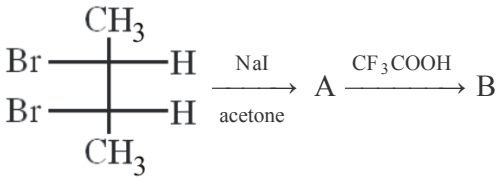
(B) 

(C) 

(D) 

Space for Rough Work

5. Select the correct statement(s) for the given reaction is/are.



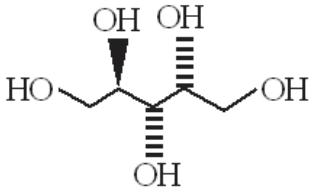
(A) 'B' is an epoxide

(B) 'A' is cis-alkene

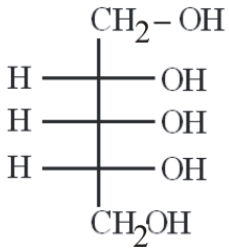
(C) 'B' does not have chiral centers.

(D) 'A' is produced via anti-elimination.

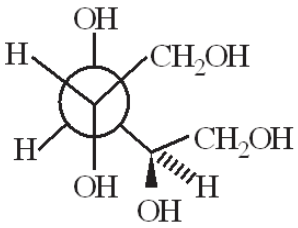
6.



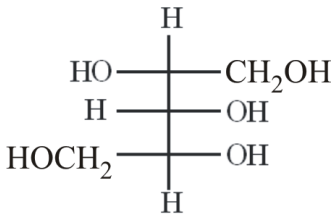
(i)



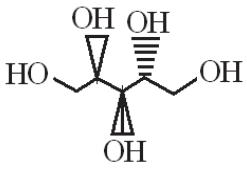
(ii)



(iii)



(iv)



(v)

(A) (i) and (iv) are enantiomers

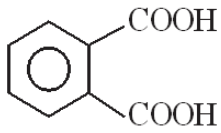
(B) (i) and (iii) are diastereomers

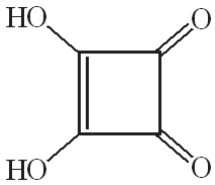
(C) (ii) and (iii) are meso compounds

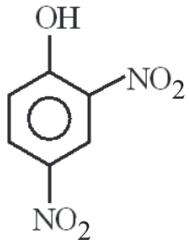
(D) (ii) and (iii) are diastereomers.

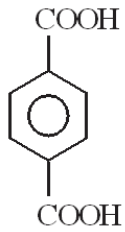
Space for Rough Work

7. Which of the following species gives CO_2 with $NaHCO_3$?

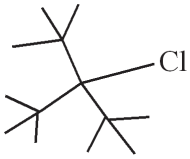
(A) 

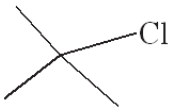
(B) 

(C) 

(D) 

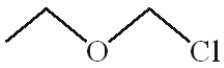
8. Select the incorrect statements

(A) Rate of hydrolysis of 

is slower than that of 

, via SN^1 mechanism.

(B) Rate of solvolysis of $(CH_3)_3CBr$ is faster in H_2O than in C_2H_5OH.

(C) 

reacts faster than 

in both SN^2 reactions.

(D) $CH_3O–CH=CH–CH_2Br$ reacts slower than $H_2C=\overset{\overset{OCH_3}{|}}{C}–CH_2Br$ in SN^1 solvolysis reaction

Space for Rough Work

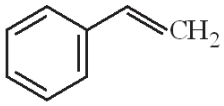
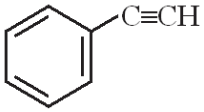
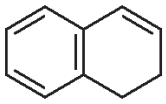
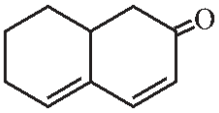
SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **TWO (02)** List-Match sets.
- Each List-Match set has **TWO** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has Four entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **Four** options are given in each Multiple Choice Question based on List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks	:	+3	If ONLY the option corresponding to the correct combination is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e., the question is unanswered)
Negative Marks	:	–1	In all other cases

Paragraph for Questions 9 and 10

In the following, List-I contains some organic compounds, and List-II contains properties.

	List-I		List-II
(I)		(P)	Aromatic compound
(II)		(Q)	More acidic than C_2H_5OH
(III)		(R)	Undergoes addition reaction with electrophiles
(IV)		(S)	Undergoes substitution reaction with odd electron species
		(T)	Produces gas with Grignard reagent.
		(U)	Soluble in NaOH

9. Which of the following options has the correct combination considering List-I and List-II.

(A) (II) – P, Q, R, S, T, U
(B) (III) – P, R
(C) (II) – P, Q, R, S, T
(D) (III) – R, S, T

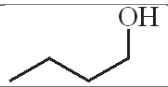
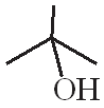
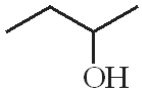
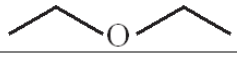
10. Which of the following options has the correct combination considering List-I and List-II.

(A) (I) – P, S, U
(B) (IV) – P, T
(C) (I) – P, S, T
(D) (IV) – S

Space for Rough Work

Paragraph for Questions 11 and 12

List-I contains four isomers of $C_7H_{10}O$ and List-II contains reactions & properties.

	List-I		List-II
(I)		(P)	Produces H_2 on reaction with metallic Na
(II)		(Q)	Undergoes oxidation to aldehyde or ketone by PCC
(III)		(R)	Cloudiness appears in Lucas Test at room temperature
(IV)		(S)	Produces CH_4 with CH_3MgBr
		(T)	forms H–bond with H_2O
		(U)	Gives red color in Victor Meyer's test.

11. Which of the following options has the correct combination considering List-I and List-II.

(A) (I) – P, Q, S, T, U

(B) (I) – P, Q, R, S, T

(C) (IV) – S, T, U

(D) (IV) – R, T, U

12. Which of the following options has the correct combination considering List-I and List-II.

(A) (II) – P, Q, R, T

(B) (II) – P, R, T, U

(C) (III) – P, Q, R, S, T

(D) (III) – Q, R, S, T, U

Space for Rough Work

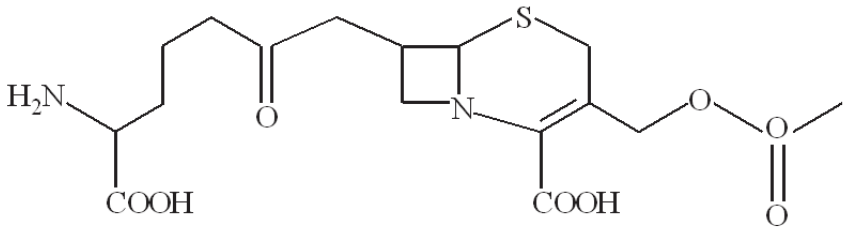
SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.
Zero Marks : 0 In all other cases.

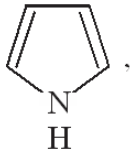
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1. Calculate the number of stereoisomers for the following compound :



2. How many of the following species are electrophilic in nature.

$:CH_2$ (Singlet), $\cdot\dot{C}H_2$ (Triplet), NO,



SO_3, $CH_3-C\equiv O^+$, $Ph-\overset{+}{N}\equiv N$, Cl_2, MnO_4^-

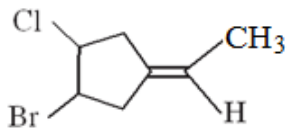
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Space for Rough Work

3. How many of the following molecules show geometrical as well as optical isomerism.

pent –2, 3–diene, octa–2, 3, 6 – trine, 1, 3–dimethyl cyclohexane

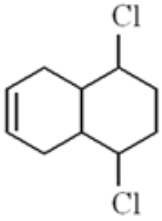
(i) (ii) (iii)



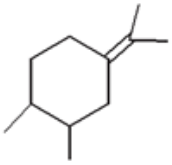
(iv)

4 - Chlorocyclooct - 1 - ene

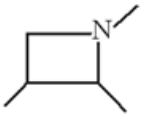
(v)



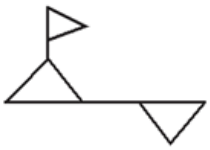
(vi)



(vii)

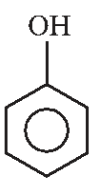


(viii)

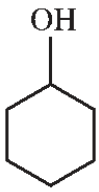


(ix)

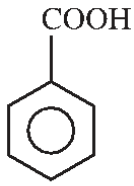
4. How many of the following compounds are soluble in aqueous NaOH.



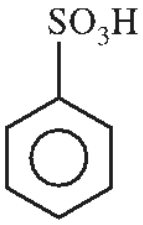
(i)



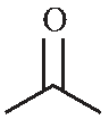
(ii)



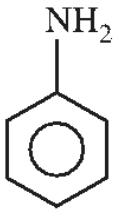
(iii)



(iv)



(v)



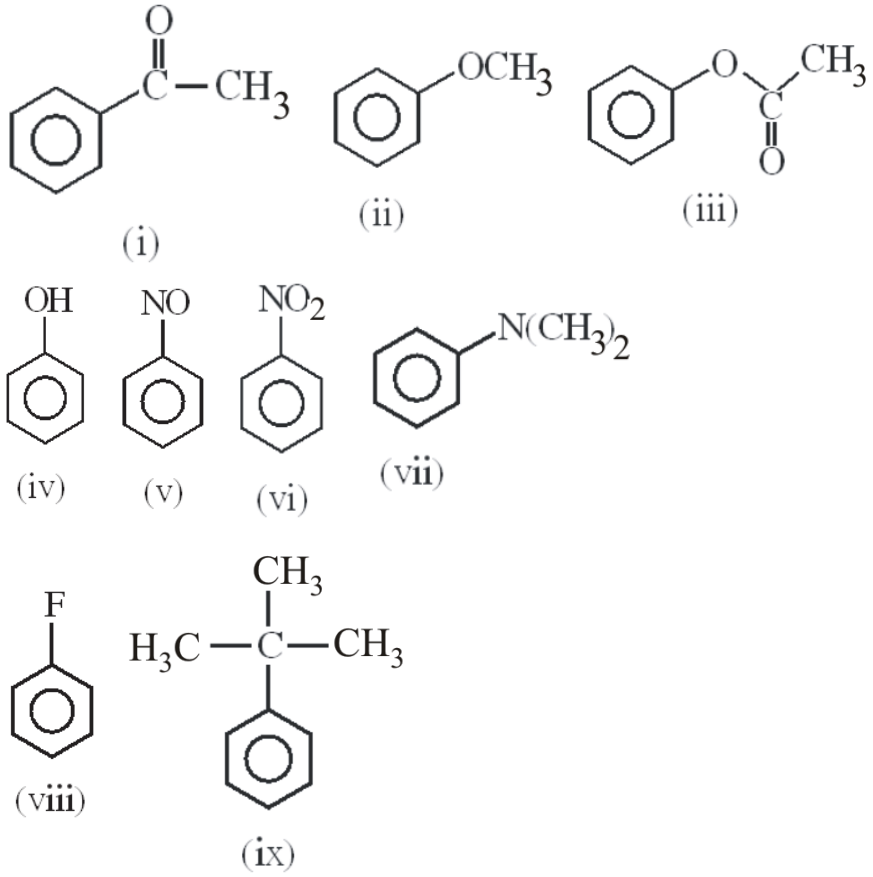
(vi)

$CHCl_3$

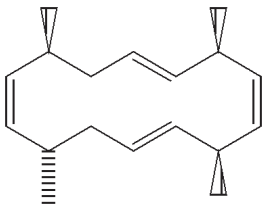
(vii)

Space for Rough Work

5. How many of the following compounds are ortho – para directing in electrophilic aromatic substitution.



6. Find the total number of pairs of enantiomers and pairs of diastereomers obtained upon complete ozonolysis of the following compound



Space for Rough Work

PART-3 : MATHEMATICS

SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : –1 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –1 marks.

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1. Let L be line containing by the planes $P_1 : x + 3y = 2$ and $P_2 : 3x - y + z = 0$. If C lies on L and $A \equiv (5, -1, -5)$, $B \equiv (1, 1, -2)$, then

(A) shortest distance between lines L and AB is $\sqrt{\frac{22}{57}}$

(B) shortest distance between lines L and AB is $\sqrt{\frac{88}{57}}$

(C) minimum area of ΔABC is $2\sqrt{\frac{319}{114}}$ sq. units

(D) minimum area of ΔABC is $\sqrt{\frac{319}{114}}$ sq. units

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Space for Rough Work

2. Consider, $f(x) = 15\sin^{-1}(\sin x) - [3 + [1 + [1 + \{x\}]\{x\}]\{x\}]\ x$

where [k] and {k} denotes greatest integer function and fractional part function of k respectively. Identify which of the following statement(s) is(are) correct?

(A) Number of solutions of the equation f (x) = 0 in $[-\pi, \pi]$ is 3

(B) Number of solutions of the equation f (x) = 0 in $[-3\pi, 3\pi]$ is 5

(C) Sum of the solutions of the equation f(x) = 0 in $[-2\pi, 3\pi]$ is $\frac{5\pi}{2}$

(D) Sum of the solutions of the equation f(x) = 0 in $[-\pi, 5\pi]$ is $\frac{5\pi}{2}$

3. If $A = \{\theta : 2\cos^2\theta + \sin\theta \leqslant 2\}$ and $B = \{\theta : \sin^{-1}(\cos\theta) \leqslant \cos^{-1}(\cos\theta)\}$ where $\theta \in [0, 2\pi]$, then

(A) number of integers in $A \cap B = 4$

(B) $\left[\frac{\pi}{4}, \frac{5\pi}{6}\right] \subseteq A \cap B$

(C) $\left[\pi, \frac{3\pi}{2}\right] \subseteq A \cap B$

(D) $A \cup B = \left[\frac{\pi}{6}, 2\pi\right]$

4. Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors such that $(|\vec{a}| - 1)^2 + \left(|\vec{b}| - 2\right)^2 + (|\vec{c}| - 3)^2 \leqslant 0$ and $\left|\vec{a} - 2\vec{b} + \vec{c}\right|^2 + \left|\vec{a} + \vec{b} - 2\vec{c}\right|^2 + \left|-2\vec{a} + \vec{b} + \vec{c}\right|^2 = 126$, then

(A) $\vec{a}, \vec{b}, \vec{c}$ constitute a triangle

(B) $\cos(\hat{a} \wedge \vec{b}) + (\cos\vec{b} \wedge \vec{c}) + \cos(\vec{c} \wedge \vec{a}) = 1$

(C) $\left(\vec{a} \times \vec{b}\right) \times \vec{c} = 3\left(2\vec{a} - \vec{b}\right)$

(D) $\left|\vec{a} + 2\vec{b} + 4\vec{c}\right| = 7$

Space for Rough Work

5. Identify which of the following statement is(are) correct on the set A = {a, b, c, d}

(A) R = {a, a), (b, b), (c, c), (d, d), (a, b), (b, a)} is an equivalence relation on A.

(B) R = {a, a), (b, b), (c, c), (d, d), (a, b)} is transitive relation on A.

(C) Number of symmetric relations on A is equal to 2^{10}.

(D) Number of reflexive relations on A is equal to 2^{14}.

6. If A(1, 1, 1) and C(9, 10, 12) are the two vertices of a triangle ABC and equation of bisector $\angle ABC$ is $\frac{x-1}{4} = \frac{z-6}{-2}$ & $y = 4$ then

(A) Equation of side BC is $\frac{x-5}{4} = \frac{y-7}{3} = \frac{z-9}{3}$

(B) Equation of side BC is $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$

(C) Co-ordinates of vertex B is (1, 5, 2)

(D) Co-ordinates of vertex B is (1, 4, 6)

7. A relation U is defined as $U = \{(x, y) : x^y = y^x ; x, y \in N\}$ relation U is

(A) Symmetric (B) Transitive (C) Reflexive (D) Void relation

8. Let $f : R^+ \to R$, $f(\lambda) = 1$, where $\lambda \in R^+$ and $f(x)\, f(y) + f\left(\frac{\lambda}{x}\right) f\left(\frac{\lambda}{y}\right) = 2f(xy) \forall\, x, y \in R^+$, then

(A) range of f(x) contains only two integers (B) f(x) is periodic function

(C) $\sum_{r=-50}^{50} f\left(\frac{51-r}{101-r}\right) = 101$, $r \in I$ (D) f(x) is bounded function

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **TWO (02)** List-Match sets.
- Each List-Match set has **TWO** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has Four entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **Four** options are given in each Multiple Choice Question based on List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks	:	+3	If ONLY the option corresponding to the correct combination is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e., the question is unanswered)
Negative Marks	:	–1	In all other cases

Paragraph for Questions 9 and 10

In ΔABC, $\begin{cases} \vec{AB} = \hat{i} + \sqrt{\alpha}\hat{j} + \alpha\hat{k} \\ \vec{AC} = \hat{i} - \sqrt{\alpha}\hat{j} + \alpha\hat{k} \end{cases}$ and area of $\Delta ABC = \Delta$.

There are some expressions are given in List-I whose values are given in List-II below.

	List-I		List-II
(I)	greatest value of $\angle BAC$	(P)	1
(II)	The length of side $\overrightarrow{BC}$, when $\angle BAC$ is maximum	(Q)	2
(III)	The least value of $\lvert\Delta/\alpha\rvert$ is	(R)	3
(IV)	The least value of $\lvert\Delta/\alpha\rvert$ occurs when α is	(S)	$\sqrt{2}$
		(T)	$\tan^{-1}(2\sqrt{2})$
		(U)	$\sin^{-1}\left(\frac{1}{3}\right)$

9. Which of the following is only the **CORRECT** combination?

(A) I → U (B) II → Q (C) III → R (D) IV → Q

10. Which of the following is only the **CORRECT** combination?

(A) I → T (B) II → R (C) III → P (D) IV → Q

Space for Rough Work

Paragraph for Questions 11 and 12

Consider the function $y = \sin^{-1}x$, $y = \tan^{-1}x$, $y = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ and $y = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$. The possible integral values of 'a' for which the graph of the function $y = |x - a|$ will cut the graph of given functions is mentioned in List-II.

	List-I		List-II
(I)	$y = \sin^{-1}x$	(P)	$a = -2$
(II)	$y = \tan^{-1}x$	(Q)	$a = -1$
(III)	$y = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$	(R)	$a = 0$
(IV)	$y = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$	(S)	$a = 1$
		(T)	$a = 2$
		(U)	$a = 3$

11. Which of the following is **CORRECT** combination?

(A) (I) → P,Q,R,S (B) (II) → P,Q (C) (I) → R,S,T (D) (II) → P,Q,R

12. Which of the following is **INCORRECT** combination?

(A) (III) → R,S,T,U (B) (IV) → R,S,T,U

(C) (III) → P,Q,R (D) (II) → Q,R,S,T

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SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.
Zero Marks : 0 In all other cases.

---

1. A rectangle ABCD of dimensions $\sqrt{85}$ and $2\sqrt{85}$ is folded along the diagonal BD such that planes ABD and CBD are perpendicular to each other. Let the position of the vertex A remains unchanged and C_1 is the new position of C. Then the distance of C_1 from A is equal to

2. The number of solution(s) to equation $2^x = x^2$, $x \in R$ is k, then fundamental period of the function $f(x) = kx - [kx]+[|\sin(\cos x)|]$ where [t] denotes greatest integer function less than or equal to t, is

3. If the range of the function $f(x) = (\sin^{-1}x)^3 + (\cos^{-1}x)^3 + \left(\sin^{-1}(4x^4 - 4x^2 + 1)\right)^3$ is $[a\pi^3, b\pi^3]$, then $\left(\frac{8a}{b}\right)$ is equal to

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Space for Rough Work

4. The maximum volume of parallelopiped having vectors along co-terminal edges as $\vec{a}\times\vec{b},\ 2\vec{b}\times\vec{c},\ 3\vec{c}\times\vec{a}$, where $\vec{a}=(\sin^2\alpha)\hat{i}+2\hat{j}-3\hat{k};\ \vec{b}=\hat{i}+2(\sin^2\alpha)\hat{j}-2\hat{k};\ \ \vec{c}=\hat{i}+\hat{j}-(\sin^2\alpha)\hat{k},\ \alpha\in R$ is

5. If 'a' is number of value(s) of 'x' satisfying $3^x + 4^x + 5^x + \ldots\ldots + n^x - (n+1)^x = 0$ where n is a natural number and $3 < n < 100$ and f is number of integers in the range of the function, $y = \sin^{-1}(\sin^{-1}x)$. Then minimum value of the expression $\left| e^a x^2 + [f-e]|x| + e \right|$; $x \in R$ is

   Note: [t] denotes greatest integer function less than or equal to t and e is napier's constant.

6. Total number of students appearing for campus placements from IIT X, IIT Y and IIT Z are 290, 340 and 360 respectively.
   Company A can only hire from IIT X and IIT Y.
   Company B can only hire from IIT Y and IIT Z.
   Company C can only hire from IIT Z and IIT X.
   If total hiring done by company A, B and C are 280, 300 and 185 respectively.
   IIT X and IIT Y secured exactly 90% and 80% placement respectively & 'N' is the number of students remains unplaced, then the value of $\frac{N}{10}$ is

Space for Rough Work

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Space for Rough Work

NAME OF THE CANDIDATE .....

FORM NO. .....

I have read all the instructions and shall abide by them. _____ Signature of the Candidate	I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same. _____ Signature of the Invigilator

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ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) path to success

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2022 - 2023)

JEE (Advanced)
PART TEST
04-12-2022

JEE(Main+Advanced) : ENTHUSIAST COURSE [PHASE : TEAS, TOAS, TAAS, TNAS, I(A) & LIVE-I(A)] (SCORE-I)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A,B,C,D	B,C,D	B,C	A	C	B,D	A,B,C	A,B,D
SECTION-I (ii)	Q.	9	10	11	12				
	A.	A	A	C	D				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	3.00	3.50 to 3.95	17.14 to 17.15	18.75	6.00	20.00		

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A	B,C,D	C,D	B,C,D	A,D	A,B,C,D	A,B,C,D	A,D
SECTION-I (ii)	Q.	9	10	11	12				
	A.	B	D	A	C				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	8.00	7.00	8.00	4.00	7.00	0.00		

PART-3 : MATHEMATICS

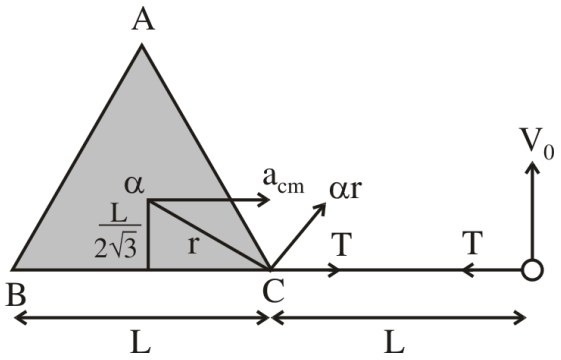
SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A,D	A,B,C,D	A,B,C	C,D	A,B,C	A,D	A,B,C	B,C,D
SECTION-I (ii)	Q.	9	10	11	12				
	A.	B	A	C	D				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	17.00	0.33	0.25	294.00	2.71 to 2.72	22.50		

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

2. Ans (B,C,D)



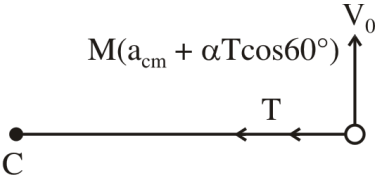
For plate: $T = M\,a_{cm}$

$$T\frac{L}{2\sqrt{3}} = I_{cm}\alpha = \frac{ML^2}{24}\alpha$$

$$\Rightarrow \quad T = \frac{ML\alpha}{4\sqrt{3}} \Rightarrow \quad Ma_{cm} = \frac{ML\alpha}{4\sqrt{3}}$$

$$\therefore \quad a_{cm} = \frac{L\alpha}{4\sqrt{3}}$$

Path of particle will be circular with respect to point C.



Acceleration of C towards right is $(a_{cm} + \alpha r\cos 60°)$.

With respect to C :

$$T + M\left(a_{cm} + \alpha\frac{L}{\sqrt{3}} \times \frac{1}{2}\right) = \frac{Mv_0^2}{L}$$

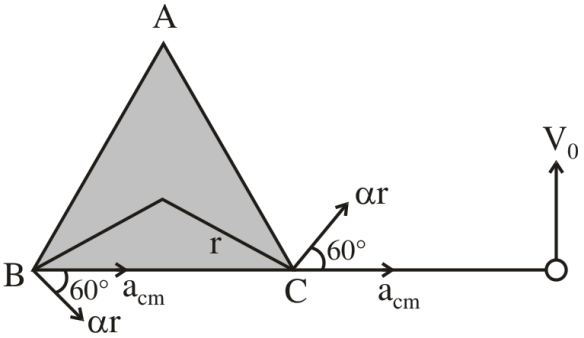
$$\Rightarrow \quad T + Ma_{cm} + \frac{M\alpha L}{2\sqrt{3}} = \frac{Mv_0^2}{L}$$

$$\Rightarrow \quad T + T + 2T = \frac{Mv_0^2}{L}$$

$$\Rightarrow \quad 4T = \frac{Mv_0^2}{L} \quad \Rightarrow \quad T = \frac{Mv_0^2}{4L}$$

Radius of curvature of path of point mass M with respect to ground just after projection

$$= R = \frac{v_0^2}{a_\perp} = \frac{v_0^2}{(T/M)} = \frac{v_0^2}{\left(\frac{v_0^2}{4L}\right)} = 4L$$



Clearly acceleration of B and C will be same in magnitude just after the projection.

3. Ans (B,C)

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} + \frac{1}{(-20)} = \frac{1}{(-25)}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{20} - \frac{1}{25} \Rightarrow \frac{1}{v} = \frac{5-4}{100} \Rightarrow v = 100$$

$$\therefore m = \frac{-v}{u} \Rightarrow m = \frac{-100}{-20} = 5$$

$\therefore$ Radius of image circle = 5 mm

Acceleration = $\sqrt{(\omega^2 R)^2 + (\alpha R)^2} = R\sqrt{\omega^4 + \alpha^2}$

$\therefore$ Acceleration $\propto$ R as ω and α will be same for object and image.

$$\therefore \frac{\text{acceleration of image}}{\text{acceleration of object}} = \frac{5}{1} = 5$$

Velocity of object = $\omega R = \alpha t \cdot R = 1 \times 2 \times 1 = 2$ mm/s

Velocity of image = $\alpha \cdot t \cdot R' = 1 \times 2 \times 5 = 10$ mm/s

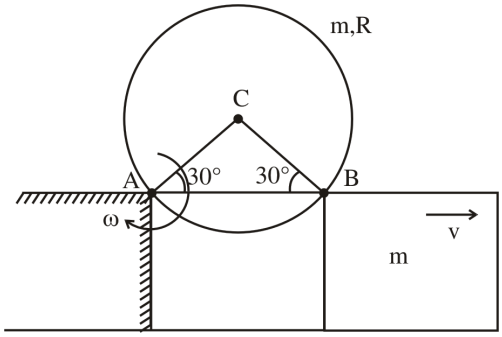
$\therefore$ Relative velocity = 10 – 2 = 8 mm/s as both velocities are in same direction.

4. Ans (A)

$v \cos 30° = \omega R\sqrt{3}\cos 60°$

$v = \omega R$

Now, using conservation of energy



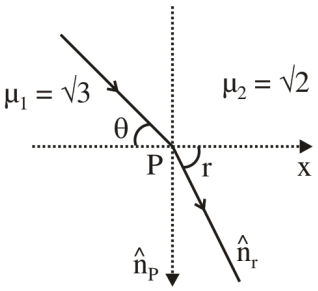
$$mgR(1 - \sin 30°) = \frac{1}{2} \times 2mR^2\omega^2 + \frac{1}{2}mv^2$$

$$\frac{mgR}{2} = \frac{3}{2}mv^2$$

$$\Rightarrow gR = 3v^2$$

$$\Rightarrow v = \sqrt{\frac{gR}{3}} = \sqrt{\frac{10 \times 0.3}{3}} = 1 \text{ m/s}$$

5. Ans (C)



Angle between normal x-axis and incident ray is

$$(5\sqrt{2}\hat{i} - 5\hat{j} - 5\hat{k}).(\hat{i}) = 10 \cdot 1 \cdot \cos\theta$$

$$\cos\theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45°$$

Applying Snell's Law at P

$\mu_1 \sin\theta = \mu_2 \sin r$

$$\sqrt{3} \cdot \frac{1}{\sqrt{2}} = \sqrt{2}\sin r;\ r = 60°$$

The unit vector along the refracted ray

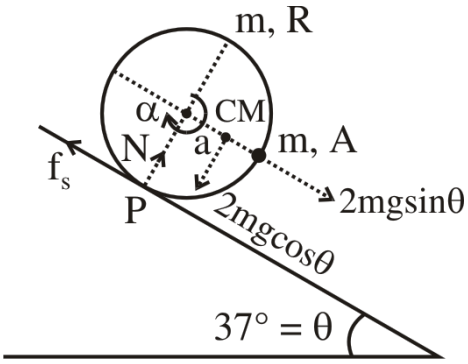
$$\hat{n}_r = \cos r\hat{i} + \sin r\hat{n}_P$$

$$\hat{n}_r = \cos 60°\,\hat{i} + \sin 60°\left(\frac{-\hat{j} - \hat{k}}{\sqrt{2}}\right)$$

$$\hat{n}_r = \frac{1}{2}\hat{i} + \frac{\sqrt{3}}{2\sqrt{2}}(-\hat{j} - \hat{k})$$

$$\hat{n}_r = \frac{\sqrt{2}\hat{i} - \sqrt{3}\hat{j} - \sqrt{3}\hat{k}}{\sqrt{8}}$$

6. **Ans (B,D)**



For pure rolling motion, a = αR

$2mg\cos\theta \times \frac{R}{2} + 2mg\sin\theta \times R = 4mR^2\alpha$

$mg(\cos\theta + 2\sin\theta) = 4m\alpha R$

$g(\cos37° + 2\sin37°) = 4\alpha R$

$g\left(\frac{4}{5} + 2 \times \frac{3}{5}\right) = 4\alpha R$

$2g = 4\alpha R \Rightarrow \alpha = \frac{g}{2R}$

$\therefore a = \alpha R = g/2$

Now, $2mg\sin\theta - f_s = 2ma$

$\frac{6mg}{5} - f_s = 2m\frac{g}{2}$

$\Rightarrow f_s = \frac{6mg}{5} - mg = \frac{mg}{5} = 2N$

Alsok, $2mg\cos\theta - N = 2m\alpha\frac{R}{2}$

$\frac{8mg}{5} - N = \frac{mg}{2}$

$\Rightarrow N = \frac{8mg}{5} - \frac{mg}{2} = \frac{11}{10}mg$

N = 11N

8. **Ans (A,B,D)**

$I_x = \frac{m(2\ell)^2}{12} + \frac{m\left(\frac{\ell}{4}\right)^2}{12} = \frac{65}{192}m\ell^2$

$I_y = \frac{m(2\ell)^2}{12} + m\left(\frac{\ell}{2}\right)^2 + \frac{m\left(\frac{\ell}{4}\right)^2}{12} + m\left(\frac{\ell}{2}\right)^2 = \frac{161}{192}m\ell^2$

$I_z = \frac{m(2\ell)^2}{6} + m\left(\frac{\ell}{2}\right)^2 + m\left(\frac{\ell}{2}\right)^2 = \frac{7}{6}m\ell^2$

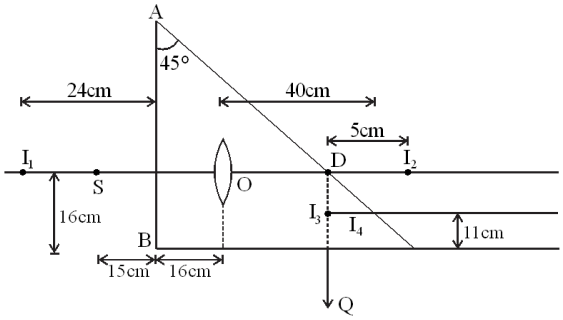
PART-1 : PHYSICS

SECTION-I (ii)

10. **Ans (A)**

Image is formed a I_1 at a distance 15 × 1.6 = 24 cm from AB

Refraction at lens



Focal length of the lens.

$\frac{1}{f} = \left(\frac{2.4}{1.6} - 1\right)\left(\frac{2}{20}\right)$ f = +20 cm

Object distance = 24 + 16 = 40 cm

Lens formula $\frac{1}{v} - \frac{1}{-40} = \frac{1}{+20}$

v = +40 cm

Image is at I_2

OD = 35 cm (from geometry of the figure)

DI_2 = 5 cm

∴ Light ray after passing through the lens will suffer total internal reflection at face AC. This is because angle of incidence is close to 45°, as we are considering paraxial rays.

$45° > \sin^{-1}\left(\frac{1}{1.6}\right)$

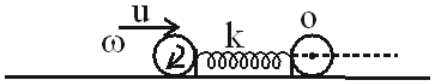
Surface AC may be treated as a mirror for which object is at I_2.

Image is formed at I_3.

DI_3 = 5 cm

Observer at P will not be able to see image as light rays suffer TIR at surface AC. Observer at Q see the image at I_4 image I_4 is above surface BC by a distance = $\frac{11}{1.6} = \frac{55}{8}$ cm

11. **Ans (C)**



Let u be the speed of A just before come in contact with the spring.

$$\therefore mgh = \frac{1}{2}mu^2 + \frac{1}{2}I\omega^2 = \frac{mu^2}{2} + \frac{I}{2}\frac{u^2}{r^2}$$

$$\Rightarrow mgh = \frac{u^2}{2}\left(m + \frac{I}{r^2}\right) \Rightarrow mgh = \frac{u^2}{2}\left(\frac{mr^2+I}{r^2}\right)$$

$$\Rightarrow u = \sqrt{\frac{2mgh\,r^2}{mr^2+I}}$$

At maximum compression both have same linear speed and as both are rolling purely being identical both have same angular speed too. Let common velocity at maximum compression be v'.

From momentum conservation $v' = \frac{u}{2}$

$= \sqrt{\frac{mgh\,r^2}{(mr^2+I)\,2}}$ {Note that net friction force on system = 0} $\omega' = \frac{\omega}{2}$

Since work done by friction is zero here, mechanical energy is conserved.

From start of compression to maximum compression :

$$\frac{1}{2}mu^2 + \frac{1}{2}I\omega^2 = 2\times\left[\frac{1}{2}mv'^2 + \frac{I}{2}\omega'^2\right] + \frac{1}{2}kx^2$$

$$= m\left(\frac{u}{2}\right)^2 + I\left(\frac{\omega}{2}\right)^2 + \frac{1}{2}kx^2$$

$$\Rightarrow \frac{1}{2}\left[\frac{mu^2}{2} + \frac{I\omega^2}{2}\right] + \frac{kx^2}{2}$$

$$\Rightarrow \frac{1}{2}mgh = \frac{kx^2}{2} \Rightarrow x = \sqrt{\frac{mgh}{k}}$$

$\therefore$ Maximum compression is same for all objects. Ratio of maximum compression in spring to maximum velocity of object A $= \frac{\sqrt{\frac{mgh}{k}}}{\sqrt{\frac{2mghr^2}{mr^2+I}}} = \sqrt{\frac{mgh}{k}\times\frac{mr^2+I}{2mgh\,r^2}}$

$$= \sqrt{\frac{(mr^2+I)}{2k\,r^2}}$$

$\therefore$ For solid sphere, ratio $= \sqrt{\frac{mr^2+\frac{2}{5}mr^2}{2k\,r^2}} = \sqrt{\frac{7m}{10k}}$

Hollow sphere $\rightarrow \sqrt{\frac{mr^2+\frac{2}{3}mr^2}{2k\,r^2}} = \sqrt{\frac{5m}{6k}}$

Ring $\rightarrow \sqrt{\frac{mr^2+mr^2}{2k\,r^2}} = \sqrt{\frac{m}{k}}$

Disc $\rightarrow \sqrt{\frac{mr^2+\frac{mr^2}{2}}{2k\,r^2}} = \sqrt{\frac{3m}{4k}}$

12. **Ans (D)**

Maximum speed of A is just before touching the spring.

$$u = \sqrt{\frac{2mgh\,r^2}{mr^2+I}}$$

Fore maximum compression $mgh = \frac{1}{2}kx^2$

$$x = \sqrt{\frac{2mgh}{k}}$$

$$\therefore \text{Ratio} = \frac{\sqrt{\frac{2mgh}{k}}}{\sqrt{2}\cdot u} = \sqrt{\frac{(mr^2+I)}{2kr^2}}$$

Putting values we get the respective answers.

PART-1 : PHYSICS

SECTION-II

1. **Ans (3.00)**

When distance of object from lens is x, let the distance of image be y.

Lens formula

$$\frac{1}{-y} - \frac{1}{-x} = \frac{1}{-f}$$

$$\frac{1}{y} = \left(\frac{1}{x} + \frac{1}{f}\right) \Rightarrow y = \frac{xf}{x+f}$$

Magnification $m = \frac{y}{x} = \frac{f}{x+f}$

Height of image is $h_i = mh_0$ (h_0 = height of object)

$$h_i = \frac{f\,h_0}{x+f}$$

Angle subtended by the image at the eye

$$\theta = \frac{h_i}{D-x+y} = \frac{f\,h_0}{(x+f)(D-x+y)}$$

Let $(x+f)(D-x+y) = Z$

$$Z = \left(D - x + \frac{xf}{x+f}\right)(x+f)$$

$$Z = -x^2 + Dx + Df$$

'Z' is maximum when

$$\frac{DZ}{dx} = 0$$

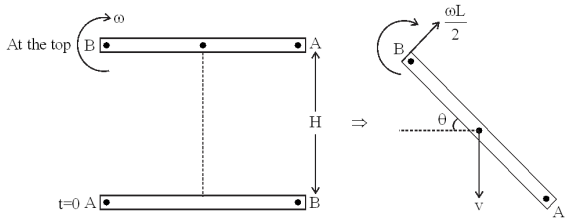
$$x = \frac{D}{2}$$

2. **Ans (3.50 to 3.95)**

Time of flight $T = \frac{2v}{g} = \frac{2\pi\sqrt{gL}}{g} = 2\pi\sqrt{\frac{L}{g}}$

$\therefore \omega = \frac{2\pi}{T} = \sqrt{\frac{g}{L}}$

As per problem rod rotates by 180° by time it's center is at top. But after this time also end B continues to rise in its vertical direction while the centre of rod would be coming down. Let θ = angle made by rod with horizontal at the instant when vB is horizontal. t be time spent after the centre of rod was at its topmost position.



$\Rightarrow \frac{\omega L\cos\theta}{2} = gt$

$\Rightarrow \sqrt{\frac{g}{L}} \cdot \frac{L}{2} \cdot \cos\left(\sqrt{\frac{g}{L}} \cdot t\right) = gt$

$\Rightarrow \cos\left(\sqrt{\frac{g}{L}} \cdot t\right) = 2t\sqrt{\frac{g}{L}}$

As per question, solution of $\cos\theta = 2\theta$ is $\theta = 0.45$

$\therefore \sqrt{\frac{g}{L}} \cdot t = 0.45$

$\Rightarrow t = 0.45\sqrt{\frac{L}{g}} = 0.45$ sec

Time taken by centre of rod to reach top $= \frac{v}{g}$

$= \frac{\pi\sqrt{gL}}{g} = \pi\sqrt{\frac{L}{g}} = \pi = 3.14$ sec

∴ Total time taken by point B to reach its topmost position = 0.45 + 3.14 = 3.59 sec.

3. **Ans (17.14 to 17.15)**

$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$\frac{1}{v} + \frac{1}{36} = \frac{1}{20} \Rightarrow \frac{1}{v} = \frac{9-5}{180}$

$\frac{1}{v} = \frac{4}{180} \Rightarrow 45$ cm

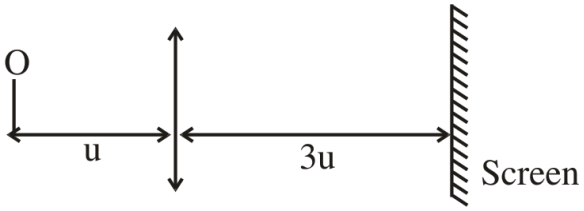
$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}; \frac{1}{20} + \frac{1}{120} = \frac{1}{f}$

$\Rightarrow \frac{1}{f} = \frac{6+1}{120}$

$\frac{1}{f} = (\mu - 1)\left[\frac{2}{R}\right] = \frac{1}{2} \Rightarrow \frac{7}{120} = \frac{1}{(4R)}$

$\Rightarrow R = \frac{120}{7} = 17.14$ to 17.15

4. **Ans (18.75)**



$\frac{h_1}{h_0} = m \Rightarrow h_1 = mh_0$

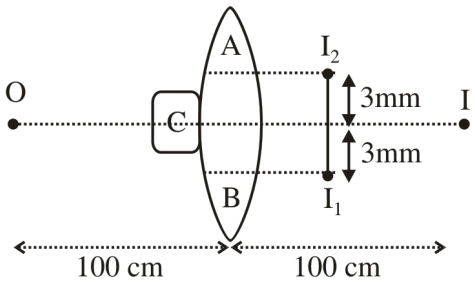
$\frac{h_1}{h_0} = \frac{1}{m} \Rightarrow h_2 = \frac{h_0}{m}$

$\frac{h_1}{h_2} = \frac{m^2}{1} = 9:1 \Rightarrow m = 3$

$4u = 100 \Rightarrow u = 25$cm

$\frac{1}{75} + \frac{1}{25}$

5. **Ans (6.00)**



The image I of point object O formed by the part C is at a distance 100 cm right on the optic axis of the lens part C.

Now,

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$$

$$\frac{1}{v}-\frac{1}{100}=\frac{1}{50}$$

$$v=+\frac{100}{3}\text{cm}$$

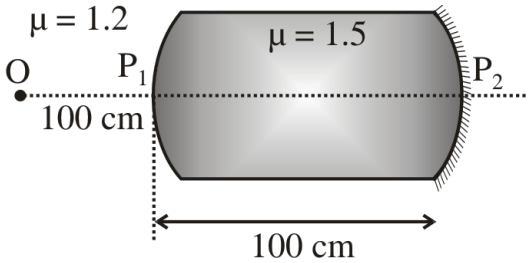
$$\therefore m=\frac{v}{u}=+\frac{1}{3}$$

$\therefore$ The distance between the two images formed by the parts A and B

d = 3 mm + 3mm = 6mm

6. **Ans (20.00)**

Refraction from P_1



$$\frac{\mu_2}{v}-\frac{\mu_1}{v}=\frac{\mu_2-\mu_1}{R}$$

$$\frac{1.5}{v_1}-\frac{1.2}{-100}=\frac{1.5-1.2}{+20}$$

$\Rightarrow v_1 = 500$ cm

Reflection from P_2

$$\frac{1}{f}=\frac{1}{v}+\frac{1}{u}$$

$$\frac{1}{-200}=\frac{1}{v_2}+\frac{1}{400},\ v_2=\frac{-400}{3}\text{cm}$$

Again refraction from P_1

$$\frac{\mu_2}{v}-\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R},$$

$$\frac{1.2}{v_3}-\frac{1.5}{100/3}=\frac{1.2-1.5}{-20}$$

$\Rightarrow v_3 = +20$ cm (left to P_1)

PART-2 : CHEMISTRY

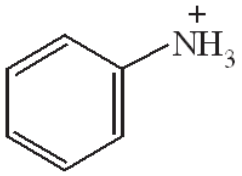
SECTION-I (i)

1. **Ans (A)**

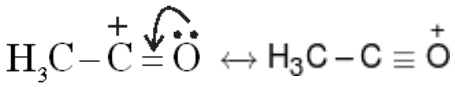
Na/liq NH_3 reduces alkyne into trans double bond.

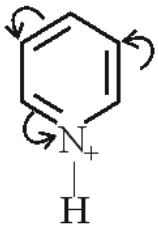
H_2/Pd, $BaSO_4$ reduces alkyne into C is double.

2. **Ans (B,C,D)**

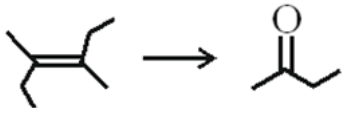
(A)  N has complete octate,

so can't accept electrons from benzene ring.

(B) 

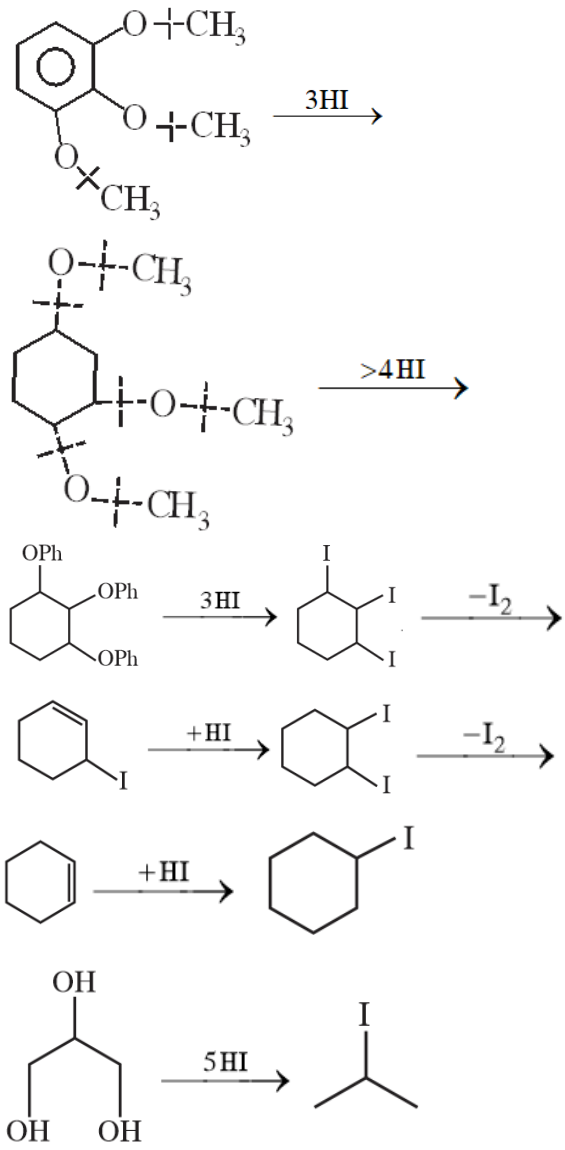
(D) 

3. **Ans (C,D)**

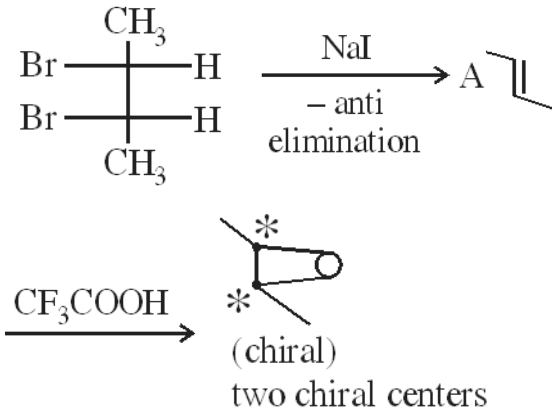


(D)
[Cis and Trans]
Optically inactive
C cannot be an alkyen

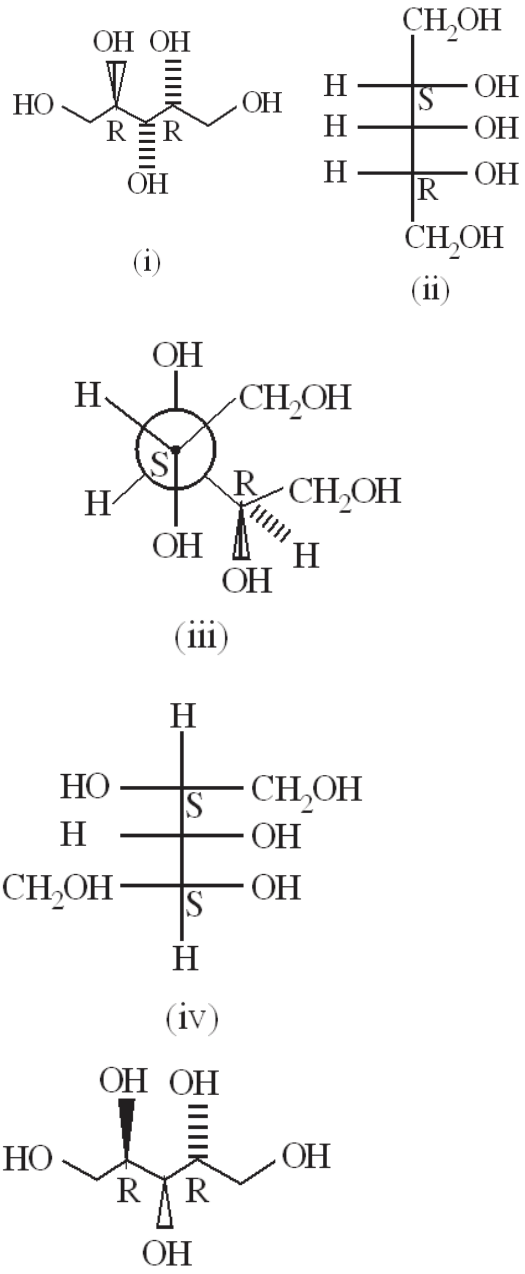
4. Ans (B,C,D)



5. Ans (A,D)



6. Ans (A,B,C,D)



i, v → identical

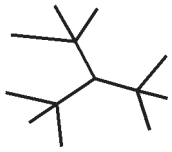
i, iv → enantiomers

ii, iii → diastereomers & meso compounds.

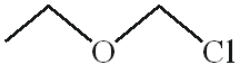
7. Ans (A,B,C,D)

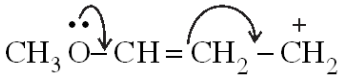
All are more acidic enough to decompose $NaHCO_3$.

8. **Ans (A,D)**

(A)  is more reactive due to large steric strain.

(B) H_2O is more ionising than ethanol, so SN^1 reactions proceeds faster in H_2O.

(C) in  , oxygen stabilies the carbocation in SN^1 & T.S in SN^2.

(D)  is more stable.

PART-2 : CHEMISTRY

SECTION-I (ii)

9. **Ans (B)**

III is aromatic and also an alkene and have no active hydrogen

II is less acidic than EtOH

10. **Ans (D)**

I is insoluble in NaOH

IV is not aromatic and have no active H.

11. **Ans (A)**

ether & 1° OH fails in Lucas Test

ether has lower boiling points then water

ether do not have active H.

12. **Ans (C)**

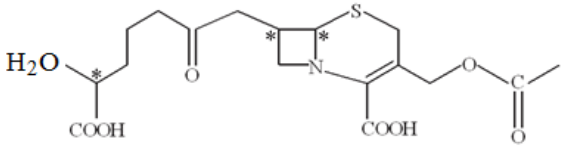
Boiling point of 1°OH > H_2O ≥ 2°OH > 3°OH.

3°OH does not undergo oxidation.

PART-2 : CHEMISTRY

SECTION-II

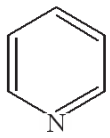
1. **Ans (8.00)**



3 chiral centers no. of stereoisomers = 8.

2. **Ans (7.00)**

$\dot{C}H_2$ triple is a radical with no vacant orbital

 is nucleophile.

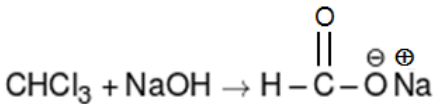
MnO_4^- with Mn^{+7} acts as an electrophile & adds to alkene.

3. **Ans (8.00)**

Can't exhibit geometrical isomerism as it is non planar.

4. **Ans (4.00)**

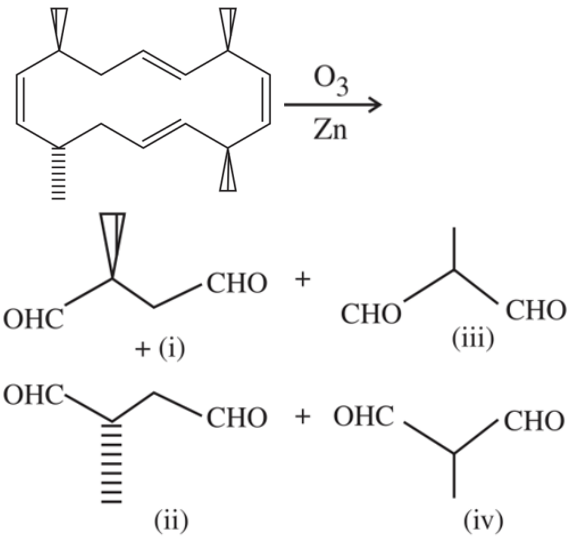
i, iii, iv, vii



5. **Ans (7.00)**

ii, iii, iv, v, vii, viii, ix

6. **Ans (0.00)**



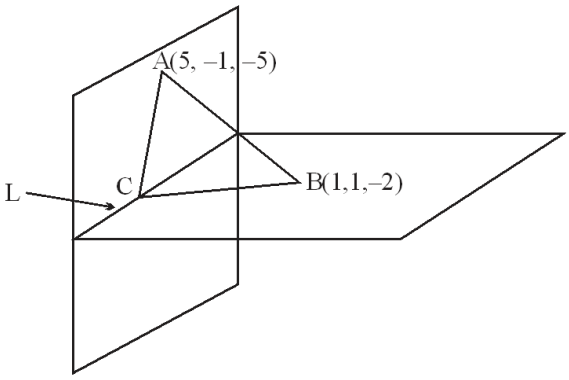
i & ii are identical

iii & iv are identical.

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,D)



$AB = \sqrt{16+4+9} = \sqrt{29}$

Shortest distance (SD) between L and AB:

where $\overrightarrow{AB} = 4\hat{i} - 2\hat{j} - 3\hat{k}$

vector along line L = $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 3 & 0 \\ 3 & -1 & 1 \end{vmatrix} = 3\hat{i} - \hat{j} - 10\hat{k}$

vector along SD = $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -2 & -3 \\ 3 & -1 & -10 \end{vmatrix} = 17\hat{i} + 3\hat{j} + 2\hat{k}$

A point on L $\equiv (2, 0, -6) \equiv P$

$\overrightarrow{BR} = \hat{i} - \hat{j} - 4\hat{k}$

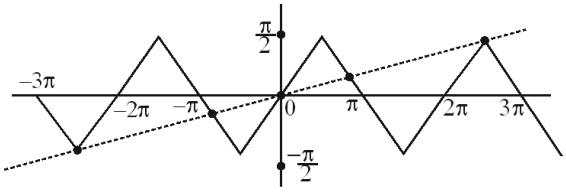
$\therefore \quad SD = \left| \dfrac{17 - 31 - 8}{\sqrt{289+961+4}} \right| = \dfrac{22}{\sqrt{1254}} = \sqrt{\dfrac{22}{57}}$

and minimum area of $\Delta ABC = \sqrt{\dfrac{319}{114}}$

2. Ans (A,B,C,D)

$5 \sin^{-1}(\sin x) - x = 0$

$\sin^{-1}(\sin x) = x/5$



Verify the options.

3. Ans (A,B,C)

$\sin\theta(2\sin\theta - 1) \geqslant 0 \Rightarrow \sin\theta \geqslant \dfrac{1}{2}$ or $\sin\theta \leqslant 0$

$A \equiv \left[\dfrac{\pi}{6}, \dfrac{5\pi}{6}\right] \cup [\pi, 2\pi] \cup \{0\}$

$\sin^{-1}(\cos\theta) \leqslant \cos^{-1}(\cos\theta)$

$\cos^{-1}(\cos\theta) \geqslant \dfrac{\pi}{4} \Rightarrow \cos\theta \leqslant \dfrac{1}{\sqrt{2}}$

$\Rightarrow B = \left[\dfrac{\pi}{4}, \dfrac{7\pi}{4}\right]$

$A \cap B = \left[\dfrac{\pi}{4}, \dfrac{5\pi}{6}\right] \cup \left[\pi, \dfrac{7\pi}{4}\right]$

$A \cup B = \{0\} \cup \left[\dfrac{\pi}{6}, 2\pi\right]$

4. Ans (C,D)

$|\vec{a}| = 1, |\vec{b}| = 2, |\vec{c}| = 3$

$\left|\vec{a} - 2\vec{b} + \vec{c}\right|^2 + \left|\vec{a} + \vec{b} - 2\vec{c}\right|^2 + \left|-2\vec{a} + \vec{b} + \vec{c}\right|^2 = 126,$

$\Rightarrow 6\left(|\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 - \vec{a}\cdot\vec{b} - \vec{b}\cdot\vec{c} - \vec{c}\cdot\vec{a}\right) = 126$

$\Rightarrow |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 - \vec{a}\cdot\vec{b} - \vec{b}\cdot\vec{c} - \vec{c}\cdot\vec{a} = 21$

$\Rightarrow \vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a} = -7$

$\Rightarrow |\vec{a} + \vec{b} + \vec{c}|^2 = \sum |\vec{a}|^2 + 2\sum \vec{a}\cdot\vec{b} = 0$

$\Rightarrow \vec{a} + \vec{b} + \vec{c} = 0$

$\Rightarrow \vec{a} + \vec{b} = -\vec{c} \Rightarrow |\vec{a} + \vec{b}|^2 = |\vec{c}|^2 \Rightarrow \vec{a}\wedge\vec{b} = 0$

$|||^{ly}\ \vec{b}\wedge\vec{c} = \vec{c}\wedge\vec{a} = \pi$

$\vec{b} = 2\vec{a}, \ \vec{c} = -3\vec{a}$

Now, verify the options.

5. Ans (A,B,C)

(A) & (B) obviously correct.

Number of symmetric relations on A such that n (A) = m is equal to $(2)^{\frac{m(m+1)}{2}} = 2^{10}$.

Number of reflexive relations on A such that n (A) = m is equal to $(2)^{m^2-m} = 2^{12}$.

Hence, options A,B,C are correct.

6. **Ans (A,D)**

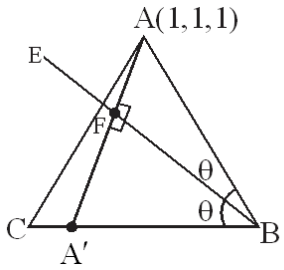


Image of A in BE lies on BC which is A' (5, 7, 9)

$\therefore$ equation of BC is $\frac{x-5}{4} = \frac{y-7}{3} = \frac{z-9}{3}$

Now point of intersection of BE and BC is (1, 4, 6).

7. **Ans (A,B,C)**

$x^x = x^x \Rightarrow$ U is reflexive

$x^y = y^x \Rightarrow y^x = x^y \Rightarrow$ U is symmetric

also U is transitive

(Because let $(x_1, y_1) \in U$ & $(y_1, z_1) \in U$, then (x_1, z_1) also belongs U)

8. **Ans (B,C,D)**

Put $x = y = 1$

$f^2(1) + f^2(1) = 2f(1)$

$f^2(1) - 2f(1) + 1 = 0 \Rightarrow (f(1) - 1)^2 = 0.$

$\Rightarrow f(1) = 1$

Now, $y = 1$ gives

$f(x) = f\left(\frac{\lambda}{x}\right) \forall x > 0$

Now put $y = \frac{\lambda}{x}$

$f(x) f\left(\frac{\lambda}{x}\right) + f\left(\frac{\lambda}{x}\right) f(x) = 2f(\lambda)$

$\Rightarrow f(x) f\left(\frac{\lambda}{x}\right) = 1$

$\therefore f^2(x) = 1 \forall x > 0$ [Since $f(x) = f(1/x)$]

If we put $x = y = \sqrt{t}$

$\Rightarrow f^2(\sqrt{t}) + f^2\left(\frac{\lambda}{\sqrt{t}}\right) = 2f(t) \Rightarrow f(x) = 1$

$\Rightarrow$ So, f (x) is periodic function, Range of f (x) = {1}.

PART-3 : MATHEMATICS

SECTION-I (ii)

10. **Ans (A)**

$\left|\overrightarrow{AB}\right| = \sqrt{1+\alpha+\alpha^2} = \left|\overrightarrow{AC}\right|$ and $\overrightarrow{BC} = -2\sqrt{\alpha}\,\hat{j}$

$\Rightarrow \left|\overrightarrow{BC}\right| = 2\sqrt{\alpha}$

$\Rightarrow \overrightarrow{AB}.\overrightarrow{AC} = 1 - \alpha + \alpha^2$

$\Rightarrow \cos A = \frac{1-\alpha+\alpha^2}{1+\alpha+\alpha^2} \leqslant 1$

$\Rightarrow 1 - \alpha + \alpha^2 \leqslant 1 + \alpha + \alpha^2 \Rightarrow \alpha \geqslant 0$

But $\frac{1-\alpha+\alpha^2}{1+\alpha+\alpha^2} \in \left[\frac{1}{3}, 1\right]$

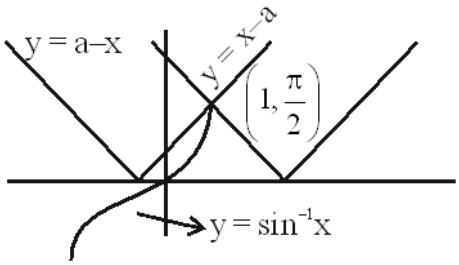
$\therefore \cos A \geqslant \frac{1}{3}$

$\Rightarrow A_{\text{greatest}}$ for $\cos A = \frac{1}{3}$

$\Rightarrow A = \tan^{-1}(2\sqrt{2})$

$\Rightarrow \left|\frac{\Delta}{\alpha}\right| = \sqrt{\alpha + \frac{1}{\alpha}} \geqslant \sqrt{2}$

12. Ans (D)



$y = \sin^{-1}x$

$y = x - a$ passes through

$\left(1, \frac{\pi}{2}\right)$, $\frac{\pi}{2} = 1 - a \Rightarrow a = 1 - \frac{\pi}{2}$

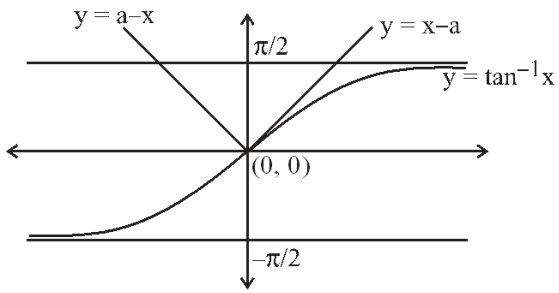
Also $y = a - x$ passes through $\left(1, \frac{\pi}{2}\right)$

$\frac{\pi}{2} = a - 1 \Rightarrow a = 1 + \frac{\pi}{2}$

So $1 - \frac{\pi}{2} \leqslant a \leqslant 1 + \frac{\pi}{2}$

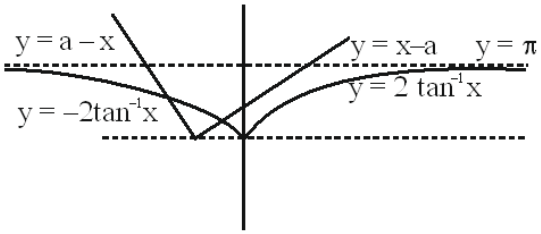
Integral values of 'a' is 0, 1, 2

(II) $y = \tan^{-1}x$



clearly $a \geqslant 0$

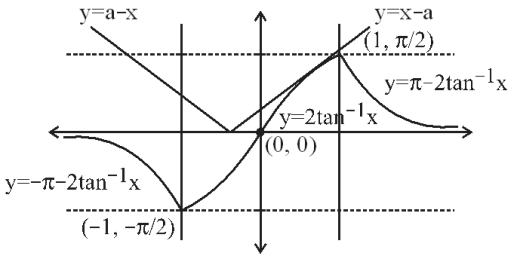
(III) $y = \cos^{-1}\left(\frac{1 - x^2}{1 + x^2}\right)$



Clearly the graph of $y = |x - a|$ cuts the graph of

$y = \cos^{-1}\left(\frac{1 - x^2}{1 + x^2}\right)$ the all values of $a \in R$.

(IV)



$y = \sin^{-1}\left(\frac{2x}{1 + x^2}\right)$

$y = x - a$ touches

$y = 2\tan^{-1}x$

$1 = \frac{2}{1 + x^2}$

$x = 1$

$y = 2\tan^{-1}(1) = \frac{\pi}{2}$

Point $\left(1, \frac{\pi}{2}\right)$ lies on $y = x - a$

$\frac{\pi}{2} = 1 - a$,

$a \geqslant 1 - \frac{\pi}{2}$, clearly integral values of a = 0, 1, 2,

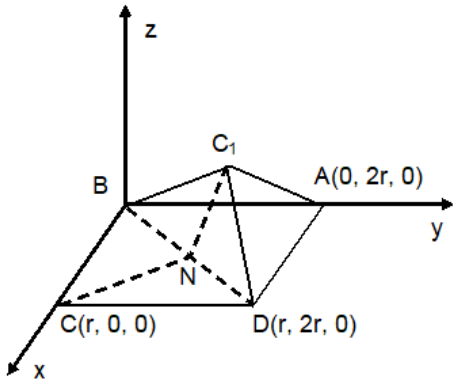
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PART-3 : MATHEMATICS

SECTION-II

1. **Ans (17.00)**

Let $r = \sqrt{85}$

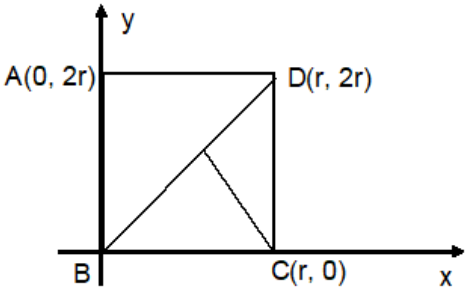


Let the rectangle ABCD initially lies in xy plane with B lying at origin BC along x-axis and BA along y-axis. Equation of BD in xy plane is $y = 2x$.

So, the coordinates of foot of the perpendicular N from C on BD are

(r/5,2r/5) and length $CN = \dfrac{2r}{\sqrt{5}}$

Clearly $CN = C_1N$



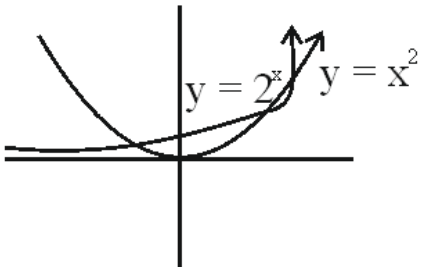
Hence the coordinates of various points in 3-D are

A (0, 2r, 0), C (r, 0, 0), D (r, 2r, 0), N (r/5, 2r/5, 0) and $C_1(r/5, 2r/5, 2r/\sqrt{5})$

Now $AC_1 = \dfrac{\sqrt{85}\ r}{5}$

$\Rightarrow AC_1 = 17$

2. **Ans (0.33)**



Number of solution of equation $2^x = x^2$ is 3

Now k = 3

$f(x)=3x-[3x]+[|\sin(\cos x)|]$

$f(x)=\{3x\}+[|\sin(\cos x)|]$

Since, $\{3x\}$ is periodic with period 1/3 and $[|\sin(\cos x)|] = 0\ \forall\ x \in R$. So, answer is 0.33.

3. **Ans (0.25)**

$$f(x)|_{min.} = f\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi^3}{32} + 0 = \frac{\pi^3}{32}$$

$$f(x)|_{max.} = f(-1) = \frac{7\pi^3}{8} + \frac{\pi^3}{8} = \pi^3$$

$$\therefore \text{Range} = \left[\frac{\pi^3}{32}, \pi^3\right]$$

$$\Rightarrow \frac{8a}{b} = \frac{1}{4} = 0.25.$$

4. **Ans (294.00)**

$$\left[\vec{a}\ \vec{b}\ \vec{c}\right]^2 = \begin{vmatrix} \sin^2\alpha & 2 & -3 \\ 1 & 2\sin^2\alpha & -2 \\ 1 & 1 & -\sin^2\alpha \end{vmatrix}^2$$

$$= \left[2\sin^6\alpha - 10\sin^2\alpha + 7\right]^2$$

Let $f(\alpha) = 2\sin^6\alpha - 10\sin^2\alpha + 7$

Let $\sin^2\alpha = p : p \in [0, 1]$

$H(p) = 2p^3 - 10p + 7$

$H'(p) = 6p^2 - 10 < 0$

$\therefore f(x) \in [-1, 7] \Rightarrow \left[\vec{a}\ \vec{b}\ \vec{c}\right]^2 = [0, 49]$

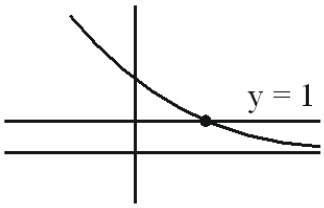
$\therefore v = 6\left[\vec{a}\ \vec{b}\ \vec{c}\right]^2 = [0, 294].$

5. **Ans (2.71 to 2.72)**

$3^x + 4^x + 5^x + \ldots\ldots n^x = (n+1)^x$

divide by $(n+1)^x$

$$\left(\frac{3}{n+1}\right)^x + \left(\frac{4}{n+1}\right)^x + \left(\frac{5}{n+1}\right)^x + \ldots\ldots\ldots + \left(\frac{n}{n+1}\right)^x = 1$$



clearly one value of 'x' satisfying the equation so a = 1.

Range of $y = \sin^{-1}\left(\sin^{-1} x\right)$ is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

So integers –1, 0, 1. So f = 3

$|ex^2 + e| = e|x^2 + 1|$ minimum value of expression is e.

6. **Ans (22.50)**

Placed student in IIT X = 261

Placed student in IIT Y = 272

Total placed by A + B + C = 280 + 300 + 185 = 765

∴ Placed student in IIT Z = 765 – 533 = 232

So number of unplaced students in

IIT'S = N = 29 + 68 + 128 = 225